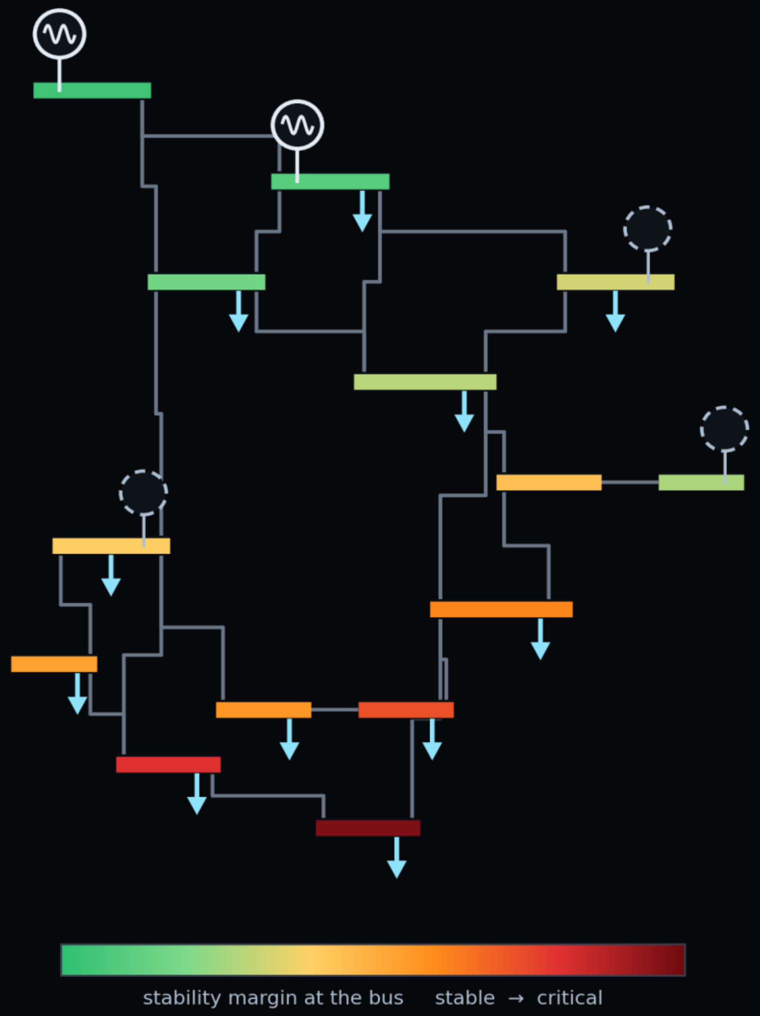
LECTURE 12.1

Power Grid Stability Estimation

The state problem: what is the grid doing right now, when you can only measure part of it. Algebraic first, then dynamic — and the physics is what fills the gaps.

Deep Learning for Engineering · Aalborg University

AC power flow · weighted least squares · the swing equation · PMU data



LEARNING OBJECTIVES

What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **write** — the power flow equations, and say why they are constraints and not a simulation
- **explain** — observability as a property of the graph, not of sensor quality
- **formulate** — estimation as a residual the network is judged against, not a map it l
- **build in** — the voltage band and the slack angle, rather than penalising them
- **identify** — inertia and damping from a recorded frequency disturbance
- **say** — where the estimate fails, and what that failure looks like

WHAT TODAY ASSUMES

from L5.1	graph networks, and message passing
from L7.1	residual, composite loss, hard enforcement
from L8.2	identifying a parameter from a curve
from L11.2	ODEs in a loss, not PDEs

TWO HALVES, TWO OPERATORS

Algebraic constraints for the steady state, then the swing equation once the grid moves. The machinery does not change.

AND REAL DATA

Energinet and Fingrid publish enough to do both halves from measurement.

From a Car You Can Hold to a Machine You Cannot

WHAT CHANGES, AND WHAT DOES NOT

- L11 ended with a model checked against an instrument you could lift onto a bench
- **the grid inverts that** — the equations are the same kind of object, the access is not
- so the question stops being 'is my model right' and becomes 'what can I know from four measurement points'
- which is exactly the shape a physics residual is good at

THE SAME MACHINERY

L7 hard constraints	reappear as the slack bus and voltage limits
L8 inverse problems	reappear as inertia and damping identification
L9 residual weighting	reappears balancing measurements against physics
L11 identifiability	reappears, and bites harder

WHAT IS GENUINELY NEW

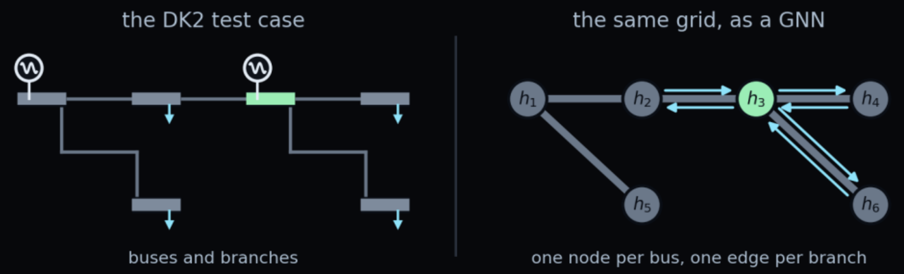
The system is a network, so the unknowns are coupled through a graph rather than through space. A missing measurement at one bus is repaired by its neighbours, and how well depends on the topology, not on distance.

That is the idea worth taking away, and it is the one the exercise is built around.

A Physics-Informed Graph Network

WHAT YOU ALREADY HAVE

- L5.1 covered graph networks alongside CNNs, and this is where they are appropriate here
- **a grid is not a grid of pixels** — the neighbours are whoever the edges connect
- so message passing, not convolution, and the adjacency comes from the topology



THE NETWORK, CONCRETELY

inputs	per-bus features, plus the adjacency
hidden	3 to 5 message-passing layers
output	voltage magnitude and angle per bus
from L5.1	permutation invariance, locality

WHY NOT AN MLP ON A FLAT VECTOR

Its input dimension is tied to one topology. A graph network takes the topology as an argument, which is what slide 14 needs.

AND THE OPERATOR IS ALGEBRAIC

No time derivative in the first half. Closer to L8.1's Poisson than to anything transient.

A Grid Is a Graph With Ohm's Law on Every Edge

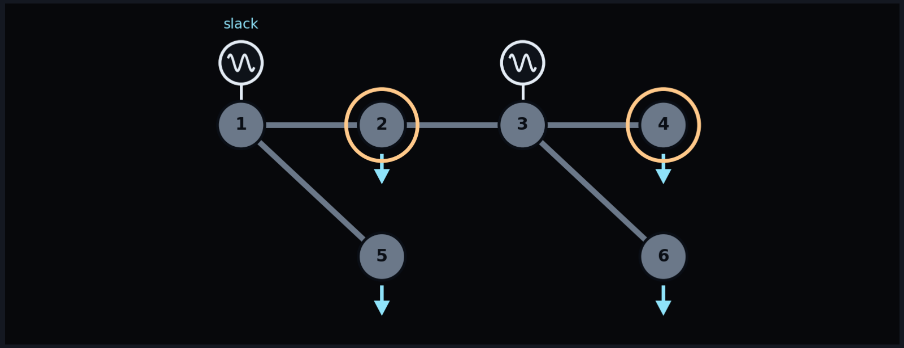
BUSES, BRANCHES, AND ONE MATRIX

- a bus is a node; a branch is an edge with an impedance
- the admittance matrix is built from topology and line parameters, both of which the operator knows
- nothing about it is learned
- **the unknowns are the voltages** — a magnitude and an angle at every bus

$$S_i = P_i + jQ_i = V_i \sum_k V_k \, Y_{ik}^* \, e^{\,j\theta_{ik}}$$

complex power injected at bus i, in terms of the voltages and the admittances

SIX BUSES IS ENOUGH TO SHOW EVERYTHING



WHY ANGLES, NOT PHASES

Only angle differences appear in any power equation, so one bus must be declared the reference and given angle zero. This is not a modelling choice you can skip: without it the problem has a one-parameter family of solutions and any optimiser will wander along it.

The Power Flow Equations

REAL AND REACTIVE INJECTION AT EVERY BUS

$$P_i = V_i \sum_k V_k \left(G_{ik} \cos\theta_{ik} + B_{ik} \sin\theta_{ik}\right)$$

$$Q_i = V_i \sum_k V_k \left(G_{ik} \sin\theta_{ik} - B_{ik} \cos\theta_{ik}\right)$$

- two equations per bus, nonlinear in the voltages and trigonometric in the angle differences
- these are constraints, not a simulation
- they hold at every instant, in steady state, whatever the disturbance history
- this is the physical content of the algebraic half of the lecture

THE STATE

$$x = \{V_i,\ \theta_i\}_{i=1}^{n}, \qquad \theta_{ik} = \theta_i - \theta_k$$

Two unknowns per bus, less the slack angle and the slack magnitude — so 2n−1 free numbers in the convention used here. Count them for your own system before you start; the count is what every observability statement refers to.

THE REFERENCE, FIXED BY HAND

$$\theta_1 = 0, \qquad V_1 = 1\ \mathrm{p.u.}$$

Enforce this the way you enforced a Dirichlet condition in L7: build it into the trial solution rather than penalising it. A slack bus that is only approximately at zero is a slack bus that is not doing its job.

What a Control Room Actually Sees

TWO MEASUREMENT SYSTEMS, DECADES APART

- **SCADA has been there since the 1960s** — injections and flows, scanned every few seconds
- phasor measurement units are the modern layer, reporting angle directly, tens of times a second
- PMUs are expensive, so they are sparse
- a real network has them at a minority of buses and a full SCADA layer underneath

SCADA

quantities	P, Q injections and flows; some $\lvert V \rvert$
rate	one scan every 2–4 s
timing	unsynchronised across substations
angle	not measured at all

PMU

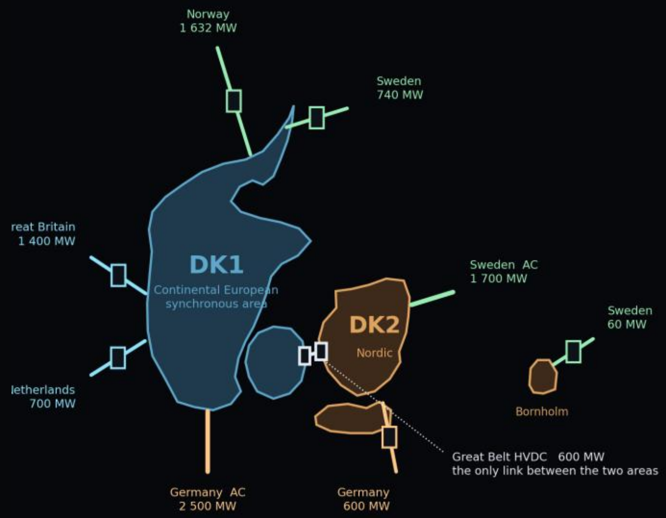
quantities	V and I phasors, magnitude and angle
rate	10–120 samples per second
timing	GPS-synchronised, microsecond class

THE CASE

Two Grids in One Small Country

DENMARK IS SPLIT, AND THAT IS USEFUL

- western Denmark is in the Continental European synchronous area; eastern Denmark is Nordic
- so two halves of one country run at different frequencies



WHAT THAT MEANS FOR THE MODEL

DK2	Nordic synchronous area, AC to Sweden
DK1	Continental area, AC to Germany
Great Belt	HVDC — power, not synchronism
Kontek	HVDC, DK2 to Germany
consequence	an HVDC link is a boundary condition

HVDC IS A BOUNDARY, NOT A BRANCH

A DC link does not appear in the swing coupling at all. It enters as a scheduled injection you impose, which is convenient: it lets you study one synchronous area without modelling its neighbours, provided you say that is what you did.

What Is Public, and What Is Not

REAL OPERATING DATA IS FREELY AVAILABLE

- Energinet publishes production, consumption and interconnector flows per price area
- Fingrid and Svenska kraftnät publish the Nordic area frequency, historically and live
- between them you have both halves of this lecture from measurement rather than simulation
- which is unusual, and worth exploiting

THE ONE THING YOU CANNOT DOWNLOAD

No TSO publishes a nodal model with line impedances — topology and parameters are withheld for security. So the network you compute on is representative, not actual, and every result carries that caveat. Say it once, clearly, and stop apologising for it.

SOURCES WORTH KNOWING

Energi Data Service	DK1/DK2 production, load, exchanges; no key
Fingrid open data	Nordic frequency, historical and live
Svenska kraftnät	the same frequency, independently
ENTSO-E Transparency	pan-European load and generation
PyPSA-Eur	an open European network topology
IEEE test cases	small, exact, for verification

Nothing here is Denmark-specific as a method. Swap in any TSO's open data and any network model — the exercise is written so the grid is an input, not an assumption.

THE PROBLEM

Fewer Measurements Than Unknowns

THE MEASUREMENT MODEL

$$z = h(x) + e, \qquad \dim z = m, \qquad \dim x = 2n - 1$$

- the measurement model is the power flow expressions plus error
- when measurement points outnumber unknowns the system is redundant and errors average down
- when they do not, it is under-determined and no amount of fitting repairs that
- **the classical answer is pseudo-measurements** — forecasts entered as if they were measurement points

THE HONEST FRAMING

A state estimator does not measure the grid. It produces the state most consistent with the readings you have and the physics you believe, and the second half of that sentence is doing at least as much work as the first.

WHY OPERATORS CARE

Everything downstream runs on the estimated state: contingency screening, dispatch, the limits shown to the operator on the wall. An error here does not stay here.

The 2003 North American blackout is the standard cautionary case — a state estimator left running on stale topology while the system moved away from what it believed.

Weighted Least Squares, and What It Costs

THE ESTIMATOR EVERY UTILITY ALREADY RUNS

$$\hat{x} = \underset{x}{\operatorname{argmin}} \sum_{j=1}^{m} \frac{\left(z_j - h_j(x)\right)^2}{\sigma_j^2}$$

- weight each residual by how much you trust that measurement point, and minimise
- solved by Gauss–Newton, fast, understood, in production for fifty years
- any new method has to beat that, not merely work
- what it needs is a well-conditioned Jacobian, and thin measurements do not give one

$$\left(H^{\mathrm{T}}RH\right)\Delta x = H^{\mathrm{T}}R\left(z - h(x)\right)$$

WHERE IT STRUGGLES

thin instrumentation	ill-conditioned, leans on pseudo-measurements
bad data	one gross error smears across the estimate
topology error	a wrong breaker status invalidates the model
fast dynamics	assumes a steady state that is not there

WHAT IT DOES WELL

With redundant, healthy instrumentation it is accurate, quick and gives a covariance you can reason about. It is the benchmark in the exercise, and beating it is not the point — understanding when it degrades is.

Report both. A method that only wins where the baseline was already weak should say so plainly.

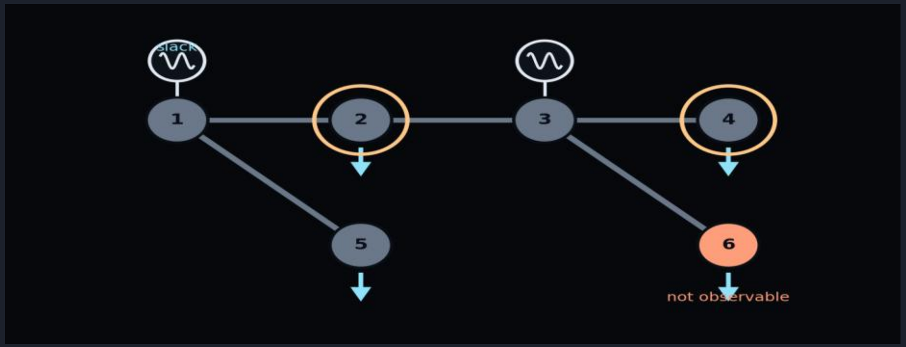
Observability Is a Property of the Graph

A RANK CONDITION, NOT A QUANTITY OF DATA

$$\operatorname{rank} H = 2n - 1 \quad \Rightarrow \quad \text{observable}$$

- observability asks whether the measurements determine the state at all
- it depends on where the measurement points are, not on how good they are
- classically the network splits into observable islands
- **a physics residual changes what is determinable** — the equations constrain the unmeasured buses too

UNOBSERVABLE, AND WHAT FILLS THE GAP



SAY WHICH IT IS

An estimate at an uninstrumented bus is an inference from the model, not a reading. The exercise asks you to report the two separately, because a plot that does not distinguish them invites the reader to trust both equally.

The Network Learns the State, Not the Physics

WHAT THE NEURAL NETWORK IS FOR

- the tempting formulation hands the network measurements and asks for the state
- the physics-informed one is different, and once seen the one used here
- the network proposes a state; the power flow equations judge it
- this is the same move as every exercise since L7

$$\mathcal{L} = \mathcal{L}_{\text{meas}} + \lambda_{\text{pf}}\mathcal{L}_{\text{pf}}$$

$$\mathcal{L}_{\text{pf}} = \frac{1}{n}\sum_i \left(\Delta P_i^{\,2} + \Delta Q_i^{\,2}\right)$$

TWO TERMS, ONE TRADE

measurement term	agree with the measurement points you have
power flow term	obey Kirchhoff everywhere, instrumented or not
λ	how much you believe the model over the measurement points
no data term	nothing here needs true states

THE WEIGHT IS THE WHOLE ARGUMENT

Push λ up and the estimate becomes a power flow solution that ignores your measurement points. Push it down and you fit noise and lose the uninstrumented buses. L9.1 made you sweep a weight and read the trade-off off a curve; do the same here, and report the curve.

The Architecture the Graph Asks For

MESSAGE PASSING, ON THE NETWORK YOU ALREADY HAVE

$$h_i^{(l+1)} = \sigma\left(W_1 h_i^{(l)} + \sum_{j \in \mathcal{N}(i)} W_2\, h_j^{(l)}\right)$$

- each bus carries a feature vector, updated from itself and its neighbours
- three properties make it the natural fit
- permutation invariance, so renumbering the buses changes nothing
- locality, matching how the physics actually couples

$$H^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(l)}W^{(l)}\right)$$

WHAT MAPS ONTO WHAT

node	a bus
node features	measurements at that bus, or zeros
edge	a transmission line
edge features	the line admittance, from Y

THE MASK MATTERS AS MUCH AS THE DATA

Most buses have no measurement. Feed zeros and the network learns that zero is a reading; feed a mask channel alongside and it learns that the value is absent. The distinction between "measured as zero" and "not measured" is the whole problem in one design choice.

What a Graph Network Adds, and What It Does Not

A CLAIM WORTH EXAMINING BEFORE YOU MAKE IT

- the usual argument is that GNNs suit grids because they embed the topology
- **true, and insufficient** — the estimator on slide 9 fits one grid with one topology
- the real case appears when you change the formulation
- train across topologies, and the model amortises rather than re-solving

$$\hat{x} = g_\theta(z,\ \mathcal{G}), \qquad \mathcal{G} = (\mathcal{V}, \mathcal{E})$$

WHERE THE GNN GENUINELY WINS

changing topology	a line out, a bus split, N-1 cases
one model, many grids	weights shared, not refitted
speed at inference	a forward pass, not an optimisation
large networks	cost grows with edges, not with n^2

WHERE IT ADDS LITTLE

one fixed grid	the residual already has the graph
single-instance fit	nothing to generalise across
six buses	message passing is overkill at this size

Build In What You Know for Certain

THE SLACK BUS AND THE VOLTAGE BAND

$$V_i = V_{\min} + (V_{\max} - V_{\min})\sigma(\hat{V}_i)$$

- voltages in a healthy network sit in a narrow band, roughly 0.94 to 1.06 per unit
- so squash the output into that band rather than penalising excursions
- **the slack angle is enforced the same way** — subtract it, do not penalise it
- both are L7.1's hard enforcement, on a grid

HARD, SOFT, AND NEITHER

slack angle	hard — subtract it, always exactly zero
voltage band	hard — sigmoid, cannot be violated
power flow	soft — a residual, weighted by λ
line limits	neither — check afterwards, do not impose

WHY LINE LIMITS ARE NOT CONSTRAINTS

A thermal limit is a rule about how the grid should be operated, not a law it must obey. Impose it and you will produce a comfortable-looking estimate of a network that is in fact overloaded — hiding the one thing the operator most needs to see.

How You Know the Estimate Is Any Good

THREE CHECKS, IN ORDER OF STRICTNESS

$$\varepsilon = \frac{\|\hat{x} - x_{\text{true}}\|}{\|x_{\text{true}}\|}$$

- on a synthetic case you know the truth, because you generated the measurements
- on real data you have none, so fall back on the residual
- and the third check is the one is often omitted
- **remove a measurement point and rerun** — a good estimator degrades gracefully

REPORT THESE SEPARATELY

instrumented buses	error where you had a reading
uninstrumented buses	error where physics alone carried it
worst bus	not the mean — the mean hides the failure
measurement point dropped	error after removing one PMU

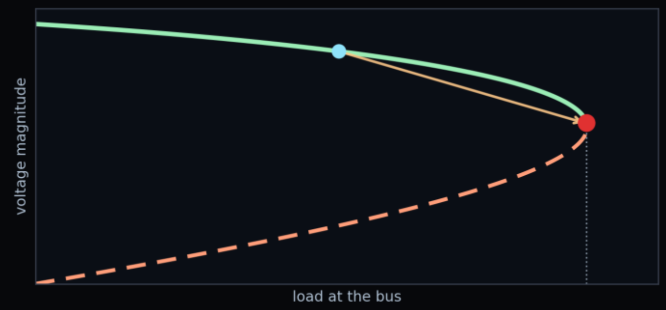
THE MEAN IS THE WRONG NUMBER

A network-wide mean error of one per cent can hide a single bus that is out by fifteen, and that bus is the one nearest the problem you were trying to see. Every table in the exercise asks for the worst case alongside the average.

Now Let the Grid Move

EVERYTHING SO FAR ASSUMED A STEADY STATE

- the power flow equations describe a grid in balance
- they are exactly true when nothing is changing, and that assumption breaks first
- a line trips, a plant disconnects, and for the next few seconds nothing is in balance
- so the second half of this lecture lets the grid move



WHAT CHANGES WHEN t APPEARS

state	voltages, plus rotor angles and speeds
equations	algebraic constraints plus ODEs
data	PMU streams, not periodic scans
unknowns	parameters too — inertia and damping
failure	not a bad number: a machine losing synchronism

The machinery does not change. This is L8.2 after L8.1 — the same problem with a time derivative restored, and the same lesson about initial conditions waiting at the end of it.

The Swing Equation

A ROTATING MASS WITH AN ELECTRICAL BRAKE

$$\frac{2H_i}{\omega_s}\ddot{\delta}_i + D_i\dot{\delta}_i = P_{m,i} - P_{e,i}$$

A synchronous generator is a heavy spinning rotor. Mechanical power comes in from the turbine, electrical power leaves through the network, and any imbalance accelerates or decelerates the rotor. That is the entire content of the equation.

H is the inertia constant — stored kinetic energy divided by rating, in seconds, typically two to eight. D lumps damping. Both are nominally known from the machine data sheet, and both drift from it in practice, which is what makes them worth estimating.

It is Newton's second law for a rotor, and it should feel like the longitudinal dynamics of L11.2 with the car replaced by a turbine. The identifiability problem is the same too: you cannot separate everything the data cannot distinguish.

AS A FIRST-ORDER SYSTEM

$$\dot{\delta}_i = \omega_i - \omega_s, \qquad \dot{\omega}_i = \frac{\omega_s}{2H_i}\left(P_{m,i} - P_{e,i} - D_i\dot{\delta}_i\right)$$

Two states per machine. This is the form you integrate, and the form the residual is written against — a second-order ODE is never what you hand an optimiser.

WHAT IS BEING IDEALISED

constant E	excitation dynamics ignored
no governor	mechanical power held fixed
classical model	no sub-transient detail
network	algebraic, infinitely fast

The Term That Ties the Machines Together

ELECTRICAL POWER OUT OF MACHINE i

$$P_{e,i} = \sum_k E_i\, E_k \left(G_{ik} \cos\delta_{ik} + B_{ik} \sin\delta_{ik}\right)$$

The electrical power depends on the angle differences between this machine and every other one. That single fact is why grid stability is a system property and not a machine property: no generator swings alone.

Note the shape of the coupling. It is sinusoidal in the angle difference, so it is restoring for small separations and stops being restoring once the difference grows past a quarter turn. A machine pushed beyond that does not come back.

Loads are folded into the network and eliminated, leaving an equivalent admittance between machine internal nodes. It is a strong simplification and the exercise says so, but it keeps the system small enough to see.

CENTRE OF INERTIA

$$\delta_{\mathrm{COI}} = \frac{\sum_i H_i\, \delta_i}{\sum_i H_i}$$

Angles are only meaningful relative to something. Referring them to the inertia-weighted mean removes the common drift and leaves the swinging you actually care about — the same act as fixing the slack angle, in the dynamic setting.

FREQUENCY, AND ITS RATE

$$f_i = f_0 + \frac{\dot{\delta}_i}{2\pi}, \qquad \mathrm{RoCoF} = \frac{d f_i}{dt}$$

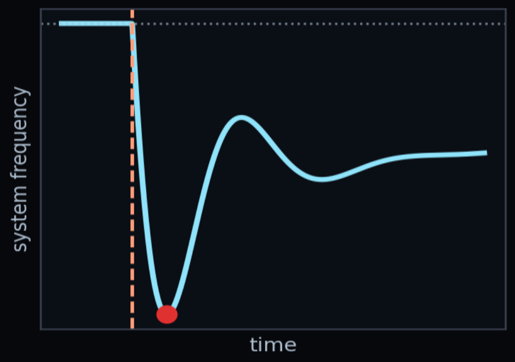
RoCoF is what protection relays watch. It is a derivative of a measured quantity, so it is noisy — and it is exactly the sort of quantity a physics-constrained estimate produces more cleanly than finite-differencing the data.

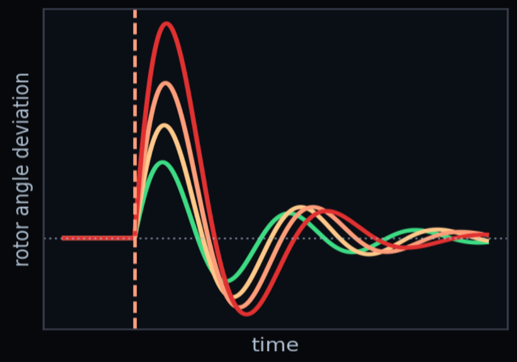
What a Real Disturbance Looks Like

THE SHAPE YOU ARE FITTING

A large unit trips. Frequency falls immediately at a rate set by the kinetic energy stored in every synchronous machine in the area — the initial RoCoF. Within seconds the fall arrests as reserves respond, and frequency settles at a new, lower value until secondary control restores it.

Three regimes, three different physics, and only the first is the swing equation on its own. Fit a window that spans all three with a model containing none of the control action and the mismatch is absorbed into your inertia estimate.





$$E_k = \sum_i H_i\, S_{n,i}, \qquad \mathrm{RoCoF}_0 = \frac{f_0\, \Delta P}{2E_k}$$

READ THE EVENT IN THIS ORDER

0 to ~1 s	inertial response — swing equation alone
1 to ~30 s	primary reserve arrests the fall
30 s onward	secondary control restores frequency
nadir	the lowest point — what protection cares about
steady error	droop, not inertia

FALLING INERTIA, IN ONE EQUATION

Initial RoCoF is the power imbalance divided by twice the stored kinetic energy. Replace synchronous plant with converter-connected generation and the denominator falls, so the same trip produces a steeper fall and less time to act. That is why this topic is current.

Dynamic State Estimation, Same Three Moves

THE NETWORK IS NOW A FUNCTION OF TIME

The state becomes a trajectory: rotor angles and speeds as functions of time, represented by a network taking t as input. Automatic differentiation supplies the time derivatives, exactly as it supplied space derivatives in L8.

The measurement term now compares against a PMU stream rather than a scan. The dynamic residual asks the trajectory to satisfy the swing equations at collocation points in time — which need not coincide with the sample instants, and generally should not.

The algebraic residual does not go away. The network equations still hold at every instant, so a full formulation carries both, and the weights between them are again yours to choose and to justify.

$$\mathcal{L}_{\text{dyn}} = \frac{1}{N}\sum_{k=1}^{N} \left\|\dot{\mathbf{x}}(t_k) - \mathbf{f}(\mathbf{x}(t_k);\lambda)\right\|^2$$

$$\mathcal{L} = \mathcal{L}_{\text{pmu}} + \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{alg}}\mathcal{L}_{\text{pf}}$$

WHY NOT A KALMAN FILTER

Extended and unscented Kalman filters are the incumbent here and they are good. They are recursive, they carry a covariance, and they run in real time.

The physics-informed formulation is not recursive: it fits a window of data at once. That is a disadvantage for real-time use and an advantage for identifying parameters, because the whole window constrains them simultaneously.

BE FAIR TO THE INCUMBENT

The exercise asks you to say what a UKF would have given on the same data and why you did not use one. A comparison you decline to make is a comparison your reader will assume you would have lost.

Inertia and Damping, Recovered From a Disturbance

PARAMETERS AS TRAINABLE VARIABLES

$$\lambda = \{H_i,\ D_i\}\ \text{trainable}$$

Declare H and D trainable and minimise the same loss. This is the inverse-diffusivity exercise of L8.2 and the traction-constant identification of L11.2, with different letters — and by now it should look routine.

It matters practically because inertia is no longer a number you can look up. As synchronous plant is replaced by converter-connected generation, the effective inertia of a region falls, varies through the day, and is not directly instrumented. Operators would like to know it and largely do not.

The disturbance is what makes it identifiable. A system sitting in steady state contains no information about H at all: nothing is accelerating, so the term multiplying H is zero. You need the grid to be disturbed, which is an uncomfortable experimental requirement.

WHAT THE DATA CANNOT SEPARATE

H and D	distinguishable only across timescales
H and P_m	a wrong mechanical power biases inertia
machines in a group	coherent machines act as one
small disturbance	H barely enters — do not trust the fit

THE L11.2 LESSON, AGAIN

An optimiser handed two parameters the data cannot distinguish will return a confident, meaningless pair. Fit the lumped quantity the data supports, report it as lumped, and say which physical parameters it contains.

L8.2 inverse α, L11.2 identification

THE CLAIM

What the Residual Contributes

THREE THINGS, AND ONLY THREE

First, estimates where there are no measurement points. The residual couples uninstrumented buses to instrumented ones through the network, so the state propagates outward. This is the headline claim and the one the exercise tests directly.

Second, noise rejection with a shape. Filtering assumes noise is fast and signal is slow; a physics residual instead assumes the trajectory satisfies a known equation, which is a much stronger and more specific prior. Derived quantities like RoCoF benefit most.

Third, parameters. A purely data-driven estimator produces a state; this produces a state and the physical constants that generated it, which is the thing an engineer can act on.

That is the list. It does not include speed, it does not include guarantees, and it does not include working when the model is wrong.

WHERE IT WINS

sparse instrumentation	the gap the method was built for
derived quantities	RoCoF, angle differences
parameter drift	inertia that is not on any sheet
offline analysis	a window, fitted at leisure

WHERE IT DOES NOT

real time	a filter is recursive, this is not
wrong topology	confidently wrong, everywhere
rich instrumentation	WLS is already fine, and faster
guarantees	no covariance falls out of it

HONESTY

How This Fails, and What It Looks Like

FOUR FAILURES YOU WILL MEET IN THE EXERCISE

THE WEIGHT IS WRONG

λ too high and the estimate is a power flow solution that ignores your measurement points; too low and you fit noise and the uninstrumented buses drift. Both look like convergence. The loss curve will not tell you — only the held-out measurement point will.

THE TOPOLOGY IS WRONG

A breaker status that does not match reality puts a wrong Y in the residual. The estimator then produces a beautifully consistent state of a network that does not exist, with small residuals throughout. This is the dangerous one.

THE DISTURBANCE IS TOO SMALL

Inertia enters through acceleration. A quiet window contains almost no information about H, and the optimiser will still return a value — one determined by your initialisation rather than by the grid.

THE MODEL IS TOO SIMPLE

The classical machine model omits excitation and governor response. Fit it to data from a real machine over a long enough window and the mismatch will be absorbed into H and D, which will come out physically implausible.

The Network You Will Compute On

REPRESENTATIVE TOPOLOGY, REAL OPERATING POINT

The exercise ships a small network whose structure follows DK2 in the ways that matter — a handful of load centres, generation concentrated at a few buses, and two HVDC links entering as fixed injections. Line parameters are plausible values, not measured ones.

Onto that structure you map real Energinet injections for a chosen hour, and real Nordic frequency for a chosen event. The topology is invented; the operating point and the dynamics are not.

This is a compromise and you should be able to defend it. The alternative — an IEEE test case with synthetic loading — is fully reproducible and entirely artificial. The alternative in the other direction is not available to anyone outside a control room.

The code takes the network as an input. Point it at an IEEE case, at a PyPSA-Eur extract, or at your own system, and everything downstream runs unchanged.

WHAT IS REAL AND WHAT IS NOT

injections	real — Energinet, per price area
frequency	real — Nordic area measurement
topology	representative, not published
line impedances	plausible, not measured
HVDC flows	real, imposed as boundary

THE HABIT THIS TEACHES

Every applied model mixes measured and assumed quantities. The discipline is not avoiding assumptions — it is labelling them, so a reader can see which of your conclusions would survive if an assumption were wrong. L10.2 asked the same of you with ESTIMATED parameters.

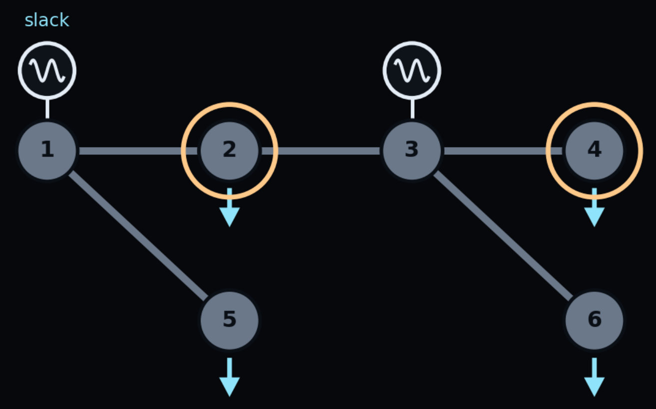
THE EXERCISE

Ex_12.1 — What You Will Build

A SIX-BUS SYSTEM, THEN A DISTURBANCE

You start from a solved power flow on a small test system, generate measurements from it with realistic noise, and hide most of them. The truth is available throughout, which is the only reason you can measure how well anything worked.

Notebook 01 is the WLS baseline. Run it first and keep its numbers: everything afterwards is reported against them, and a method that cannot beat a fifty-year-old estimator on its own ground needs to explain itself.



THE NOTEBOOKS

00 system check	the test case and its power flow, read-only
01 WLS baseline	the reference estimator — has TODOs
02 algebraic PINN	state from sparse measurement points — has TODOs
03 dynamic PINN	the swing window — has TODOs
04 inertia	the inverse problem — has TODOs
05 compare	tables, curves and the report questions

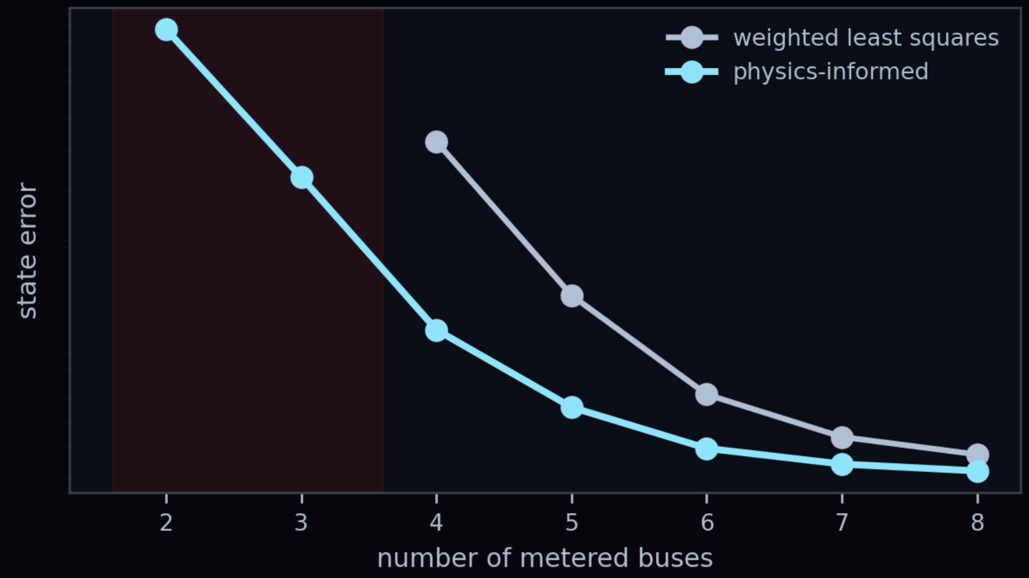
THE QUESTION THAT CARRIES THE MARKS

Notebook 05 asks which of your numbers you would show an operator, and which you would not. As in L9.2, the marks are for saying precisely what would have to be true first.

Roughly What the Numbers Look Like

ORDER OF MAGNITUDE, NOT A TARGET

On a small well-instrumented case, WLS and the physics-informed estimator agree closely and WLS is far quicker. That is the expected result and it is not a disappointment — it is the control that tells you your implementation is correct.



WHAT MUST HOLD

full instrumentation	both methods accurate; WLS faster
half the measurement points	the gap opens, in the expected direction
uninstrumented bus	error larger than at instrumented buses
quiet window	inertia estimate untrustworthy — say so
wrong topology	small residuals, wrong answer

If any of these five comes out backwards, something is wrong with the implementation and not with the physics. That is the debugging checklist.

indicative, six-bus case

OVER TO YOU

Questions

TEN TO TEST WHETHER TODAY LANDED

- why are the power flow equations constraints rather than a simulation?
- what is in the admittance matrix, and which part of it is learned?
- what does observability depend on, and what does it not depend on?
- why does a physics residual change the observability ?
- why hard-enforce the voltage band instead of penalising excursions?
- how do you check an estimate when you have no ground truth?

AND FOUR MORE

seven	why is DK2's frequency not Denmark's?
eight	what does a GNN provide that a per-case solve does not?
nine	what does the swing equation add that power flow cannot say?
ten	weighted least squares has run for fifty years. Why change?

THE ONE WITH NO SETTLED ANSWER

Where would you put the next measurement point, and how would you defend the choice?

BEFORE L12.2

Ex_12.1. Bring your estimate and its residual map.

Where Should the Next Measurement Point Be Placed?

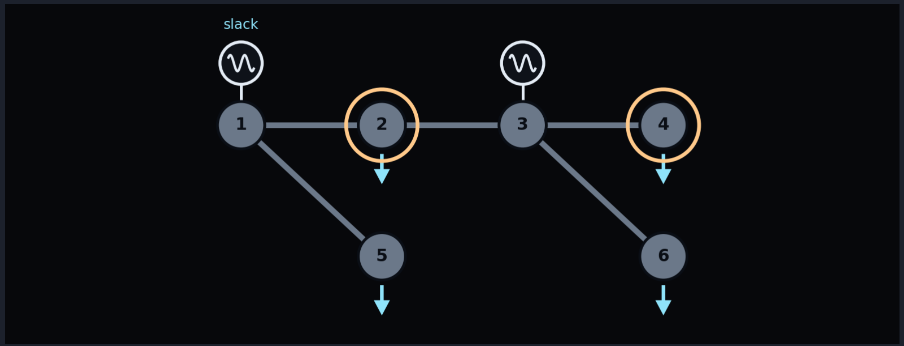
THE QUESTION THE METHOD LETS YOU ASK

Once you can estimate with sparse instrumentation, the natural question inverts: given a budget for one more PMU, where does it do the most good? That is a design question, and it is answerable by experiment — remove and add measurement points, and watch the worst-bus error.

The answer is rarely the busiest bus. It is usually the one that breaks an unobservable island, or the one furthest from existing coverage in the graph sense, and it depends on topology in ways that are hard to guess and easy to test.

This is an optional strand in the exercise and the most open-ended thing in it. It is also the closest the course comes to letting the model inform a decision rather than describe a state.

TRY IT BY EXPERIMENT



AN HONEST OPTIMISATION

Six buses means you can simply try every placement and rank them. Say that you did it exhaustively — a brute-force answer that is complete beats a clever one that might have missed the optimum, and at this size there is no excuse for the clever one.

THE POINT

One Line to Remember

A physics residual turns "I have no measurement point there" into "I have a weaker measurement there" — and the whole engineering task is knowing how much weaker.

AND THREE THINGS THAT FOLLOW FROM IT

An estimate at an uninstrumented bus is an inference. Label it as one, every time, in every plot you hand to somebody who did not run the code.

The weight between measurement and physics is not a hyperparameter to tune away. It is the statement of how much you trust your model relative to your instruments, and it belongs in the report as a number with a justification.

A model you cannot check is not a measurement. The topology failure on slide 18 is the whole argument in one example.

WHAT L12.2 DOES WITH THIS

This lecture recovers the state now. The next one asks whether the system is heading somewhere it should not — the same equations integrated forward, with the disturbance you care about absent from the training data.

That last clause is the whole difficulty, and it is where a physics residualearns most clearly.

SOURCES

References and Further Reading

THE CLASSICAL SIDE

Schweppe & Wildes (1970), Power system static-state estimation, IEEE Trans. PAS — the paper that started the field.

Abur & Expósito (2004), Power System State Estimation: Theory and Implementation — the standard reference for WLS, observability and bad-data detection.

Kundur (1994), Power System Stability and Control — the swing equation, machine models and transient stability.

THE PHYSICS-INFORMED SIDE

Raissi, Perdikaris & Karniadakis (2019), J. Comput. Phys. 378, 686–707 — the formulation used throughout this course.

Liu (2025), PINN with Python — the implementation conventions the exercises follow.

The power-system PINN literature is young and moves quickly; treat any specific accuracy claim in it as provisional until you have reproduced it.

READ IN THIS ORDER

if you are new to grids	Kundur ch. 11, then Abur ch. 1–2
if you know grids	Raissi et al., then the exercise
for the exercise	the run guide at the top of the core module
for L12.2	Kundur ch. 13, transient stability

Nothing in this lecture requires a paper you cannot get through the university library, and nothing in the exercise requires anything beyond torch, numpy and matplotlib.

NEXT

L12.2 — Prediction of Power Grid Stability

Estimation asked what the grid is doing. Prediction asks where it is going — and whether it comes back. Same equations, integrated forward, with the disturbance you care about missing from every training set you have.

BEFORE THE NEXT SESSION

run	Ex_12.1 notebooks 00 and 01
bring	your WLS baseline numbers
think about	where you would put one more PMU
read	Kundur ch. 13 if you can

The baseline matters. Everything in L12.2 is reported against something, and if you arrive without a number to compare to, you will spend the session generating one.